[Metric]

MIL-T-12014B(EA)
14 January 1981
SUPERSEDING
MIL-T-12014A
30 December 1952

MILITARY SPECIFICATION

TRIBUTYLAMINE, NORMAL (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of tributylamine.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations.

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1078 - Distillation Range of Volatile Organic Liquids.
D1193 - Reagent Water.
D1744 - Water in Liquid Petroleum Products by Karl Fischer Reagent.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical and physical characteristics. The tributylamine shall conform to the chemical and physical characteristics of table I when tested as specified therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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TABLE I. Chemical and physical characteristics

Characteristic	Requirement	Test paragraph
Color	To pass test	4.2.4.1
Purity, minimum percent by weight	97	4.2.4.2
Specific gravity, 20°/20°C	0.77 to 0.80	4.2.4.3
Boiling range, minimum percent by weight distilled between 196° and 216°C	95	4.2.4.4
Water, maximum percent by weight	0.10	4.2.4.5

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the tributylamine produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 125 milliliters (ml) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers selected in accordance with 4.2.2.1 shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Container closure gasket missing or incorrect
(f) Incorrect torque or plug closure for container
(g) Palletization not as specified (when required)
(h) Marking incorrect, missing, or illegible

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TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 250 ml. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Color. The color of the specimen shall be no darker than that of a standard containing 75 parts per million of potassium dichromate, when compared in 50-ml Nessler tubes.

4.2.4.2 Purity.

(a) Indicator solution. Add 3 parts of 0.1 percent alcoholic bromcresol green to 1 part of 0.2 percent alcoholic methyl red indicator.

(b) Procedure. Weigh to the nearest milligram 1.0 to 1.5 grams (g) of the specimen from a 10-ml weighing buret or a Lunge weighing bottle into an Erlenmeyer flask containing 45.0 ml of 0.5N hydrochloric acid. Add 50 to 75 ml of water and 4 drops of the indicator solution prepared in (a). Shake thoroughly and titrate the excess hydrochloric acid with 0.5N sodium hydroxide. The color change is from red to yellow to green. Calculate the tributylamine content as follows:

$$\text{Percent tributylamine} = \frac{18.53\ (VN - BM)}{W}$$

where: B = Milliliters of sodium hydroxide solution used,
M = Normality of sodium hydroxide solution,
N = Normality of hydrochloric acid solution,
V = Milliliters of hydrochloric acid solution used, and
W = Weight of specimen in grams.

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4.2.4.3 Specific gravity. Determine the specific gravity at 20°C by any method accurate to two decimal places.

4.2.4.4 Boiling range. Determine the boiling range according to ASTM D1078, with the addition that a volume reading shall be taken when the thermometer reading is 196°C and again when it reads 216°C.

4.2.4.5 Water. Determine the water content according to ASTM D1744, except that glacial acetic acid shall be used in lieu of methanol throughout the test.

5. PACKAGING

5.1 Unit packing. Unit packing shall be level A or C as specified (see 6.2).

5.1.1 Level A. A quantity of 206 (+2 or -0) liters of tributylamine shall be unit packed in a steel drum. The drum shall be clean and dry within and shall conform to Department of Transportation (DOT) Specification 5B. The closures shall be furnished with leak proof gaskets and pilfer-deterrent seals which shall neither affect nor be affected by the tributylamine or environment. The plug closures shall be torqued closed to within a range as specified by the drum manufacturer.

5.1.2 Level C. A quantity of 206 (+2 or -0) liters of tributylamine shall be unit packed to provide adequate protection against deterioration and leakage from the supply source to the first receiving activity for immediate use. Unit packing shall be in accordance with DOT regulations and the common carrier rules applicable to the mode of transportation.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Tributylamine unit packed level A as specified in 5.1.1 shall require no further protection for shipment.

5.2.2 Level C. Tributylamine shall be packed to assure carrier acceptance and safe delivery to the first domestic destination. Packs shall be in compliance with rules and regulations as specified in 5.1.2.

5.3 Unitization. When specified (see 6.2) packs of tributylamine shall be palletized in accordance with MIL-STD-147

5.4 Marking. Unit packs, packs, and pallet loads of tributylamine shall be marked in accordance with DOT regulations, the applicable regulations of other regulatory agencies, and MIL-STD-129. In addition, each unit pack, pack, and pallet load shall be marked to show the lot or batch number and date of manufacture of the tributylamine.

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6. NOTES

6.1 Intended use. The material covered by this specification is intended for use as a stabilizer for chemical agents.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of unit packing required (see 5.1).
(c) Level of packing required (see 5.2).
(d) If palletization is required (see 5.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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